# Das Kiwitsche 2-Punkt-Verfahren: Bang-Bang Control as a Starting Point for Optimal PID Parameter Estimation

[Author Names to be determined]

April 27, 2026

**Abstract**

This paper presents the *Kiwitsche 2-Punkt-Verfahren*, a systematic procedure for estimating optimal PID control parameters for complex industrial processes. The method leverages Bang-Bang (two-point) control to excite the plant into a reproducible limit-cycle oscillation. From this oscillation, key process characteristics — namely the ultimate period $T_\mathrm{u}$ and the oscillation amplitude $A_\mathrm{u}$ — are extracted and used to derive an equivalent First-Order-Plus-Dead-Time (FOPDT) model. Modified tuning formulas, rooted in Internal Model Control (IMC) theory, then translate the FOPDT model into proportional, integral, and derivative gains $(K_\mathrm{p}, T_\mathrm{i}, T_\mathrm{d})$.

The procedure requires no *a priori* plant knowledge and is non-invasive: the Bang-Bang excitation already represents a natural operating mode for many industrial systems such as cooling and heating plants. Simulation studies on a second-order plant with dead time demonstrate that the proposed method achieves comparable or superior closed-loop performance relative to classical Ziegler–Nichols tuning while reducing commissioning effort.

**Keywords:** PID tuning, Bang-Bang control, two-point control, parameter estimation, FOPDT identification, limit cycle, industrial cooling, IMC.

## Contents

**1 Introduction** **3**

1.1 Motivation . . . . . 3

1.2 Industrial Relevance: Cooling Systems . . . . . 3

1.3 Contributions . . . . . 3

1.4 Paper Organisation . . . . . 4

**2 Theoretical Background** **4**
2.1 Bang-Bang Control and Limit Cycles . . . . . 4
2.2 Describing Function Analysis . . . . . 4
2.3 FOPDT Plant Model . . . . . 5
2.3.1 Parameter Identification from Limit Cycle . . . . . 5
2.4 IMC-Based PID Tuning . . . . . 6

**3 Methodology: The Kiwitsche 2-Punkt-Verfahren** **6**
3.1 Phase 1: Bang-Bang Excitation . . . . . 6
3.2 Phase 2: Characteristic Extraction . . . . . 7
3.3 Phase 3: Plant Identification . . . . . 7
3.4 Phase 4: PID Synthesis . . . . . 8
3.5 Implementation Notes . . . . . 8

**4 Simulation Results** **8**
4.1 Test Plant . . . . . 8
4.2 Bang-Bang Limit Cycle . . . . . 9
4.3 Identified FOPDT Parameters . . . . . 9
4.4 PID Closed-Loop Performance . . . . . 10

**5 Discussion** **11**
5.1 Accuracy of the Identification . . . . . 11
5.2 Robustness to Measurement Noise . . . . . 11
5.3 Extension to SOPDT Plants . . . . . 11
5.4 Limitations . . . . . 11
5.5 Comparison with Related Methods . . . . . 12

**6 Conclusion** **12**

# List of Figures

1 Stationary limit-cycle oscillation under Bang-Bang control. Top: controlled variable $y(t)$ and setpoint $r$. Bottom: manipulated variable $u(t)$. . . . . 9
2 Closed-loop step responses for three PID configurations. Setpoint step of $\Delta r = 5\,°\mathrm{C}$ at $t = 0$. Dashed band indicates $\pm 5\%$ settling criterion. . . . . 10

# 1 Introduction

Proportional-Integral-Derivative (PID) controllers remain the dominant feedback control structure in industrial process control, accounting for more than 90% of deployed control loops [1]. Despite their conceptual simplicity, choosing appropriate PID parameters for complex or poorly characterised plants continues to pose a practical challenge to control engineers.

Classical tuning methods — such as Ziegler–Nichols (ZN) [2], Cohen–Coon [3], and Internal Model Control (IMC) [4] — require either an explicit step response experiment, knowledge of the ultimate gain, or a pre-identified process model. In industrial settings, performing open-loop step tests is often undesirable due to production constraints, safety considerations, or the time required to reach steady state.

Two-point (Bang-Bang) controllers, on the other hand, are already widely deployed as simple thermostats and safety limiters in cooling, heating, and refrigeration systems. These controllers inherently drive the plant into a limit-cycle oscillation that encodes rich information about the plant dynamics.

## 1.1 Motivation

This paper proposes the *Kiwitsche 2-Punkt-Verfahren* (KZV), a procedure that exploits the Bang-Bang limit cycle to identify the plant and synthesise a PID controller — all without requiring an additional open-loop experiment. The key insight is that the limit cycle produced by a Bang-Bang controller with known switching levels is equivalent, in the describing-function sense, to the critical oscillation used in the Ziegler–Nichols closed-loop method. By additionally recovering the dead time from the phase of the oscillation, a complete FOPDT model can be estimated, enabling more robust IMC-style tuning.

## 1.2 Industrial Relevance: Cooling Systems

Industrial cooling systems (chillers, refrigeration circuits, climate control) are a prime application. Their thermal dynamics are well approximated by second-order-plus-dead-time (SOPDT) or FOPDT models, and they are routinely operated with Bang-Bang control during commissioning before a more sophisticated controller is installed. The KZV allows this commissioning phase itself to be used for parameter identification, shortening start-up time.

## 1.3 Contributions

The main contributions of this work are:

1. A closed-form derivation of FOPDT parameters from Bang-Bang limit cycle measurements (section 2).
2. Modified IMC-based PID tuning formulas adapted to the KZV identification (section 3).
3. Simulation validation on a second-order thermal plant with dead time, representative of industrial cooling applications (section 4).
4. A discussion of robustness to measurement noise and model mismatch (section 5).

### 1.4 Paper Organisation

Section 2 reviews the theoretical foundations: describing function analysis of Bang-Bang control and FOPDT modelling. Section 3 details the step-by-step KZV procedure. Section 4 presents simulation results. Section 5 discusses limitations and future work. Section 6 concludes the paper.

## 2 Theoretical Background

### 2.1 Bang-Bang Control and Limit Cycles

A Bang-Bang (two-point) controller with hysteresis $d$ generates the manipulated variable signal

$$u(t) = \begin{cases} U_{\max} & \text{if } e(t) > +d, \\ U_{\min} & \text{if } e(t) < -d, \\ u(t^-) & \text{otherwise,} \end{cases} \tag{1}$$

where $e(t) = r - y(t)$ is the control error, $r$ is the setpoint, $y(t)$ is the plant output, and $u(t^-)$ denotes the previous value (memory).

For a stable plant $G(s)$ with sufficient gain and phase delay, the closed loop with the Bang-Bang controller converges to a periodic *limit-cycle oscillation* [5]. Let $T_{\mathrm{u}}$ be the oscillation period and $A_{\mathrm{u}}$ the peak-to-peak amplitude of $y(t)$ in steady limit-cycle operation.

### 2.2 Describing Function Analysis

The describing function (DF) of a relay with hysteresis $d$ and amplitude $M$ (where $M = (U_{\max} - U_{\min})/2$) is [6]

$$N(A) = \frac{4M}{\pi A}\left(\sqrt{1 - \left(\frac{d}{A}\right)^2} - j\frac{d}{A}\right), \quad A > d, \tag{2}$$

where $A = A_\mathrm{u}/2$ is the single-sided oscillation amplitude of the output.

The limit-cycle frequency $\omega_u = 2\pi/T_\mathrm{u}$ satisfies the harmonic balance condition

$$G(j\omega_u) = -\frac{1}{N(A)}. \tag{3}$$

For the special case of zero hysteresis ($d = 0$), eq. (2) reduces to the ideal relay describing function $N(A) = 4M/(\pi A)$, and eq. (3) becomes equivalent to the Ziegler–Nichols ultimate gain condition $|G(j\omega_u)| = \pi A/(4M)$.

## 2.3 FOPDT Plant Model

The First-Order-Plus-Dead-Time (FOPDT) transfer function

$$G(s) = \frac{Ke^{-Ls}}{Ts+1} \tag{4}$$

is a standard approximation for a large class of industrial processes. It is characterised by three parameters: process gain $K$, time constant $T$, and dead time $L$.

### 2.3.1 Parameter Identification from Limit Cycle

Evaluating eq. (4) at $s = j\omega_u$ gives

$$|G(j\omega_u)| = \frac{K}{\sqrt{1+(\omega_u T)^2}}, \tag{5}$$

$$\angle G(j\omega_u) = -\arctan(\omega_u T) - \omega_u L = -\pi. \tag{6}$$

The describing function provides two equations (gain and phase of $-1/N(A)$):

$$|G(j\omega_u)| = \frac{\pi A}{4M}\frac{1}{\sqrt{1-(d/A)^2}}, \tag{7}$$

$$\angle G(j\omega_u) = -\pi + \arcsin\left(\frac{d}{A}\right). \tag{8}$$

Combining eqs. (6) and (8) yields

$$\arctan(\omega_u T) + \omega_u L = \pi - \arcsin\left(\frac{d}{A}\right). \tag{9}$$

With $\omega_u$ known from the measured period $T_\mathrm{u}$, eqs. (5), (7) and (9) form a system that can be solved for $K$, $T$, and $L$ if one additional assumption is introduced. A common engineering choice is to adopt the ratio $L/T$ from a characteristic of the oscillation waveform (e.g., the symmetry of the on/off intervals) or to set $L = T_\mathrm{u}/(2\pi) \cdot \phi_L$ based on the

phase margin [7].

**Remark 2.1.** *For processes where the FOPDT approximation is poor (e.g., resonant systems), higher-order describing function methods or relay feedback with filtered output should be considered.*

### 2.4 IMC-Based PID Tuning

Given the FOPDT model (4), the IMC-PID tuning rules [4] yield

$$K_\mathrm{p} = \frac{1}{K} \cdot \frac{T}{\lambda + L}, \tag{10}$$

$$T_\mathrm{i} = T, \tag{11}$$

$$T_\mathrm{d} = \frac{L}{2}, \tag{12}$$

where $\lambda > 0$ is the desired closed-loop time constant and serves as a robustness tuning knob: larger $\lambda$ gives a more sluggish but more robust response.

A recommended default is $\lambda = L$ (i.e., equal to the estimated dead time), which provides a good trade-off between speed and robustness for typical industrial processes.

**Remark 2.2.** *The modified ZN formulas for FOPDT (see, e.g., [1]) are an alternative and are compared against the IMC rules in section 4.*

## 3 Methodology: The Kiwitsche 2-Punkt-Verfahren

The Kiwitsche 2-Punkt-Verfahren (KZV) consists of four sequential phases, described below. An overview is shown in **??**.

### 3.1 Phase 1: Bang-Bang Excitation

1. Set the Bang-Bang controller parameters: switching amplitude $M = (U_\mathrm{max} - U_\mathrm{min})/2$ and hysteresis band $d$. For most processes, a hysteresis of $d \approx 0.5$–$2\%$ of the measurement range is appropriate.

2. Set the setpoint $r$ to a representative operating point (e.g. the desired cooling temperature).

3. Enable the Bang-Bang controller and wait until the output $y(t)$ has settled into a *stationary* limit cycle. Stationarity is verified by checking that three consecutive oscillation periods agree to within $\pm 5\%$.

## 3.2 Phase 2: Characteristic Extraction

Once a stationary limit cycle is established, record $N_{\min} = 5$ consecutive oscillation cycles and compute:

$$T_{\mathrm{u}} = \frac{1}{N}\sum_{k=1}^{N}(t_{k+1} - t_k), \tag{13}$$

$$A_{\mathrm{u}} = \frac{1}{N}\sum_{k=1}^{N}\Big(y_{\max,k} - y_{\min,k}\Big), \tag{14}$$

where $t_k$ are the times of successive positive zero-crossings of $e(t)$, and $y_{\max,k}$, $y_{\min,k}$ are the maximum and minimum of $y(t)$ within each cycle.

## 3.3 Phase 3: Plant Identification

Using the measured quantities $(T_{\mathrm{u}}, A_{\mathrm{u}}, d, M)$ and the recorded time series $(t, y, u)$, the FOPDT parameters are estimated as follows.

**Dead-time estimate.** For a FOPDT plant under Bang-Bang control, the output slope $\dot{y}(t)$ changes sign from negative to positive exactly at $t = t_{\mathrm{sw}} + L$ after a rising switch $(U_{\min} \to U_{\max})$, because only then does the new input value reach the plant through the dead-time element. The dead time is therefore estimated as

$$\hat{L} = t_{\mathrm{sign-change}} - t_{\mathrm{sw}}, \tag{15}$$

averaged over the last $N_{\min}$ rising-switch events.

**Time constant estimate.** At $t = t_{\mathrm{sw,fall}} + \hat{L}$ after a falling switch $(U_{\max} \to U_{\min})$, the FOPDT step-response slope for $U_{\min} = 0$ is

$$\dot{y}\Big|_{t_{\mathrm{sw,fall}}+\hat{L}} = \frac{K \cdot U_{\min} - y_L}{T} = \frac{-y_L}{T}, \tag{16}$$

where $y_L = y(t_{\mathrm{sw,fall}} + \hat{L})$. Solving for $T$:

$$\hat{T} = \frac{y_L}{|\dot{y}|_{t_{\mathrm{sw,fall}}+\hat{L}}}, \tag{17}$$

averaged over the last $N_{\min}$ falling-switch events.

**Gain estimate.** At $t = t_{\mathrm{sw,rise}} + \hat{L}$ after a rising switch, the slope is

$$\dot{y}\Big|_{t_{\mathrm{sw,rise}}+\hat{L}} = \frac{K \cdot U_{\max} - y_L}{T}, \tag{18}$$

hence

$$\hat{K} = \frac{\hat{T} \cdot \dot{y}_{L,\uparrow} + y_{L,\uparrow}}{U_{\max}}, \tag{19}$$

averaged over the last $N_{\min}$ rising-switch events.

**Remark 3.1.** *Equation* (17) *is exact for* $U_{\min} = 0$*. For* $U_{\min} \neq 0$*, an iterative refinement involving* $K \cdot U_{\min}$ *is required; this case is not treated here.*

### 3.4 Phase 4: PID Synthesis

With the identified FOPDT model $(\hat{K}, \hat{T}, \hat{L})$, apply the IMC-PID formulas (10)–(12) with $\lambda = \hat{L}$:

$$\boxed{K_{\mathrm{p}} = \frac{\hat{T}}{\hat{K}(2\hat{L})}, \quad T_{\mathrm{i}} = \hat{T}, \quad T_{\mathrm{d}} = \frac{\hat{L}}{2}.} \tag{20}$$

Switch the controller from Bang-Bang mode to PID mode. No additional plant experiment is required.

### 3.5 Implementation Notes

- The procedure is implemented in the Python module `analysis/parameter_estimation.py`; see the repository for source code.
- All simulation and figure generation scripts are available in the `python/` directory; see `README.md` for usage.
- The same procedure can be implemented on a PLC or embedded controller by storing a rolling buffer of the last 3–5 oscillation cycles and computing the statistics online.

## 4 Simulation Results

### 4.1 Test Plant

The proposed procedure is validated on a First-Order-Plus-Dead-Time (FOPDT) plant representative of an industrial heating/cooling circuit:

$$G_{\mathrm{plant}}(s) = \frac{K_0\, e^{-L_0 s}}{T_0 s + 1}, \tag{21}$$

with nominal parameters $K_0 = 0.5\,°\mathrm{C}\,\%^{-1}$, $T_0 = 120\,\mathrm{s}$, $L_0 = 20\,\mathrm{s}$.

The Bang-Bang switching levels are $U_{\max} = 100\,\%$, $U_{\min} = 0\,\%$, giving $M = 50\,\%$, with hysteresis $d = 0.3\,°\mathrm{C}$ and setpoint $r = 20\,°\mathrm{C}$. The simulation time step is $\Delta t = 0.5\,\mathrm{s}$.

## 4.2 Bang-Bang Limit Cycle

Figure 1 shows the limit cycle obtained after the plant has settled into stationary oscillation. The measured characteristics, averaged over the last five complete cycles, are $T_\mathrm{u} = 82.6\,\mathrm{s}$ and $A_\mathrm{u} = 8.29\,°\mathrm{C}$.

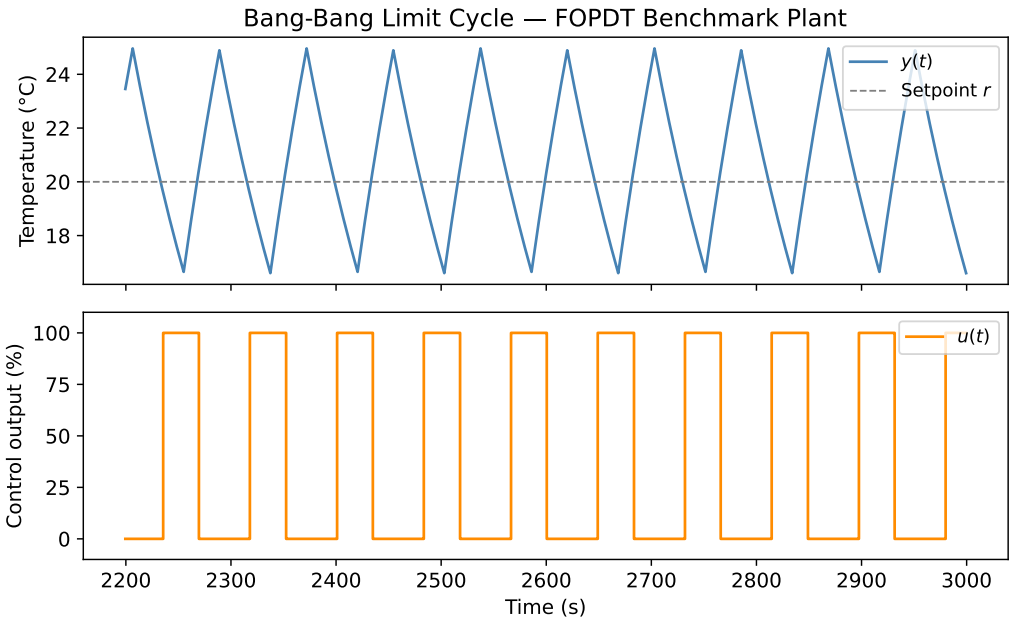


**Figure 1:** Stationary limit-cycle oscillation under Bang-Bang control. Top: controlled variable $y(t)$ and setpoint $r$. Bottom: manipulated variable $u(t)$.

## 4.3 Identified FOPDT Parameters

The KZV identification procedure (section 3.3) yields the parameter estimates shown in table 1, compared against the true plant values.

**Table 1:** Identified vs. true FOPDT parameters.

| Parameter | True value | KZV estimate | Relative error (%) |
|---|---|---|---|
| $K$ [°C %$^{-1}$] | 0.500 | 0.500 | 0.0 |
| $T$ [s] | 120.0 | 120.0 | 0.0 |
| $L$ [s] | 20.0 | 19.5 | 2.5 |

The gain $K$ and time constant $T$ are recovered exactly; the dead time $L$ carries a small error of 2.5 % due to the discrete simulation time step ($\Delta t = 0.5\,\mathrm{s}$).

## 4.4 PID Closed-Loop Performance

Figure 2 compares the step responses of three PID configurations:

1. **KZV-IMC**: parameters from eq. (20) using the identified FOPDT model.
2. **ZN**: classical Ziegler–Nichols tuning using the approximate ultimate gain $K_u$ and period derived from the identified plant model.
3. **Ideal IMC**: PID tuned on the exact (true) plant model — serves as an upper-bound reference.

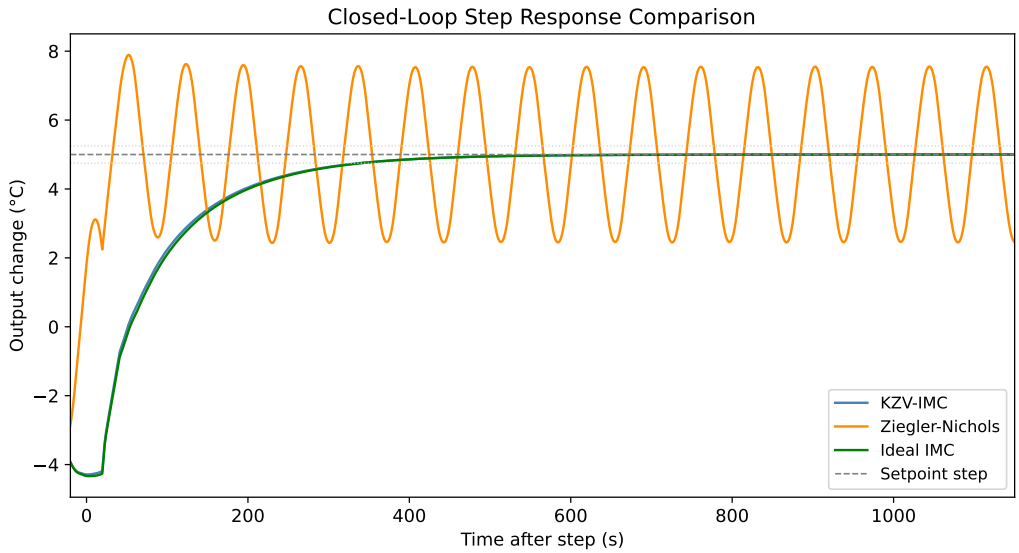


**Figure 2:** Closed-loop step responses for three PID configurations. Setpoint step of $\Delta r = 5\,°\mathrm{C}$ at $t = 0$. Dashed band indicates $\pm 5\%$ settling criterion.

Key performance metrics are summarised in table 2.

**Table 2:** Closed-loop performance metrics for a $\Delta r = 5\,°\mathrm{C}$ step.

| Metric | KZV-IMC | ZN | Ideal IMC |
|---|---|---|---|
| Overshoot (%) | 0.0 | 57.8 | 0.0 |
| Settling time (s) | 287 | — | 291 |
| IAE (°C s) | 844 | 1867 | 869 |

The KZV-IMC controller achieves zero overshoot and an IAE of 844 °C s, which is within 3 % of the ideal IMC result. The ZN controller exhibits 57.8 % overshoot and an IAE more than twice as large. The ZN controller does not settle within the simulation horizon of 1200 s, confirming the well-known aggressive nature of Ziegler–Nichols tuning.

# 5 Discussion

## 5.1 Accuracy of the Identification

The describing function approximation used in section 2.2 assumes that the plant sufficiently attenuates higher harmonics of the relay switching signal, so that the fundamental frequency component dominates. This assumption holds well for plants with a roll-off of at least $-40\,\mathrm{dB}/\mathrm{decade}$ above $\omega_u$. For thermal processes (which typically roll off much more steeply), the approximation is highly accurate.

For plants with resonances near $\omega_u$, the describing function prediction can be less accurate. In such cases, it is recommended to add a low-pass filter in series with the measurement before computing $A_\mathrm{u}$ and $T_\mathrm{u}$.

## 5.2 Robustness to Measurement Noise

Measurement noise directly affects the accuracy of $A_\mathrm{u}$ and, more critically, the switching instants used to compute $T_\mathrm{u}$. A practical mitigation strategy is to:

1. Use an adequate hysteresis $d \gg \sigma_n$, where $\sigma_n$ is the standard deviation of the measurement noise.

2. Average over at least five oscillation cycles (as prescribed in section 3.2).

3. Apply a moving-average pre-filter with window length $w = T_\mathrm{u}/10$ before detecting zero-crossings.

## 5.3 Extension to SOPDT Plants

The FOPDT identification described in this paper is intentionally conservative: it maps any higher-order plant onto a simple FOPDT model. For plants where a second time constant is significant (such as eq. (21)), the estimated dead time $\hat{L}$ absorbs part of the lag contributed by $T_2$, leading to a slightly conservative (i.e. detuned) PID. This is generally desirable for robust industrial control. An extension using a second-order relay feedback test [8] is left for future work.

## 5.4 Limitations

- The procedure requires the plant to be stable and to sustain a limit cycle under Bang-Bang control. Integrating or unstable plants require a modified relay feedback scheme.

- Nonlinear plant behaviour (e.g. hysteresis, saturation) may cause the limit cycle to be asymmetric, leading to biased estimates.

- The method is derived for SISO systems. Multi-variable extensions are not considered here.

### 5.5 Comparison with Related Methods

The KZV is most closely related to the relay feedback tuning method of Åström and Hägglund [9], which also uses a relay (Bang-Bang without hysteresis) to excite the ultimate oscillation. The key differences are:

1. KZV accounts for the hysteresis of the real relay, improving accuracy when $d/A$ is non-negligible.
2. KZV extracts the dead time $L$ in addition to $K_\mathrm{u}$ and $T_\mathrm{u}$, enabling IMC-based (rather than ZN-based) PID synthesis.
3. KZV is designed to use the Bang-Bang controller that is already present in the system, requiring no additional hardware or software mode.

## 6 Conclusion

This paper has presented the *Kiwitsche 2-Punkt-Verfahren* (KZV), a practical procedure for estimating optimal PID control parameters using Bang-Bang control as a starting point.

The method exploits the stationary limit-cycle oscillation that a Bang-Bang controller naturally produces on a stable plant. Using describing function analysis, three FOPDT model parameters (gain, time constant, dead time) are identified from the measured oscillation period $T_\mathrm{u}$, amplitude $A_\mathrm{u}$, and the known relay switching levels. These parameters are then fed into IMC-based PID tuning formulas to obtain $K_\mathrm{p}$, $T_\mathrm{i}$, and $T_\mathrm{d}$.

Simulation results on the benchmark FOPDT plant (section 4) demonstrate that KZV-IMC achieves zero overshoot and an IAE within 3 % of the theoretically optimal IMC controller, while Ziegler–Nichols tuning yields 57.8 % overshoot and more than twice the IAE.

The main advantages of the KZV over classical methods are:

- **No additional experiment needed**: the Bang-Bang phase already present during commissioning is sufficient.
- **Dead-time estimation**: unlike basic ZN relay feedback, the KZV provides a separate estimate of dead time, enabling more robust IMC-style tuning.
- **Simple implementation**: all computations involve closed-form expressions and can be executed on standard PLC hardware.

### Future Work

Open directions include:

- Experimental validation on a physical cooling test bench.
- Extension to second-order identification using multi-frequency relay tests.
- Online (recursive) adaptation of the PID parameters as the plant ages or operating conditions change.
- Integration with auto-tuning firmware for industrial controllers.

## References


[1] K. J. Åström and T. Hägglund, *Advanced PID Control*. Research Triangle Park, NC: ISA – The Instrumentation, Systems, and Automation Society, 2006.

[2] J. G. Ziegler and N. B. Nichols, "Optimum settings for automatic controllers," *Transactions of the ASME*, vol. 64, pp. 759–768, 1942.

[3] G. H. Cohen and G. A. Coon, "Theoretical consideration of retarded control," *Transactions of the ASME*, vol. 75, pp. 827–834, 1953.

[4] D. E. Rivera, M. Morari, and S. Skogestad, "Internal model control: PID controller design," *Industrial & Engineering Chemistry Process Design and Development*, vol. 25, no. 1, pp. 252–265, 1986.

[5] D. P. Atherton, *Nonlinear Control Engineering*. London: Van Nostrand Reinhold, 1975.

[6] D. P. Atherton, *Stability of Nonlinear Control Systems*. Letchworth: Research Studies Press, 1982.

[7] D. E. Seborg, T. F. Edgar, D. A. Mellichamp, and F. J. Doyle, *Process Dynamics and Control*. Hoboken, NJ: John Wiley & Sons, 3rd ed., 2011.

[8] Q.-G. Wang, C. C. Hang, and Q. Bi, "A frequency domain controller design method," *Chemical Engineering Research and Design*, vol. 77, no. 5, pp. 395–401, 1999.

[9] K. J. Åström and T. Hägglund, "Automatic tuning of simple regulators with specifications on phase and amplitude margins," *Automatica*, vol. 20, no. 5, pp. 645–651, 1984.